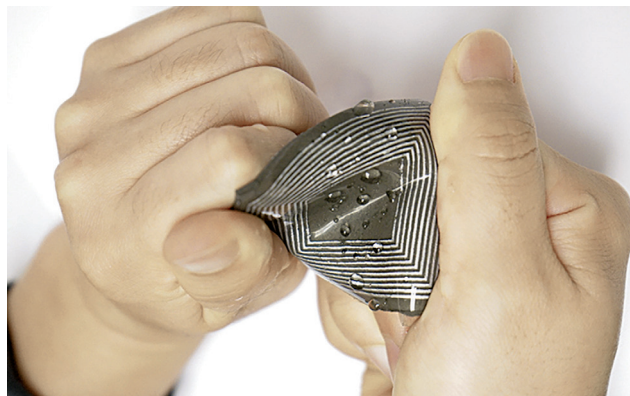
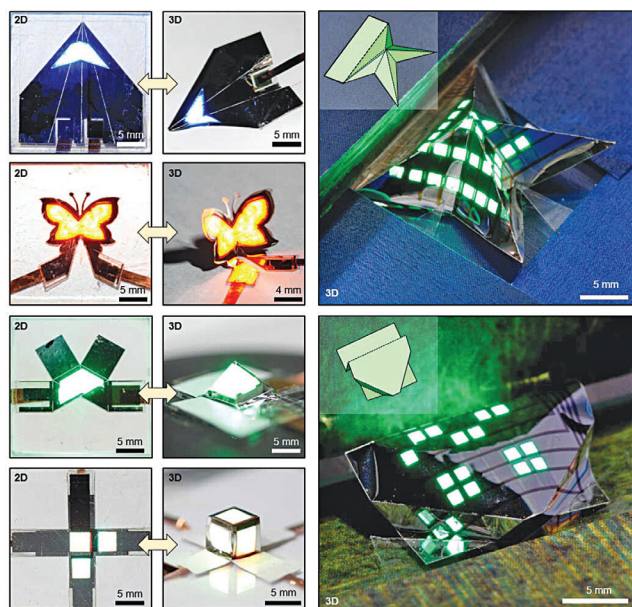


Human body motions can now power devices

A team of bioengineers at the UCLA Samueli School of Engineering has invented a novel soft and flexible self-powered bioelectronic device. The technology converts human body motions, from bending an elbow to subtle movements such as a pulse on one's wrist, into electricity that could be used to power wearable and implantable diagnostic sensors. The researchers discovered that the magnetoelastic effect, which is the change of how much a material is magnetised when tiny magnets are constantly pushed together and pulled apart by mechanical pressure, can exist in a soft and flexible system—not just one that is rigid. As the magnetic field's strength shifts, electricity is generated.



UCLA-designed self-powered, stretchable, waterproof magnetoelastic generator for bioelectronics (Credit: <https://samueli.ucla.edu>)



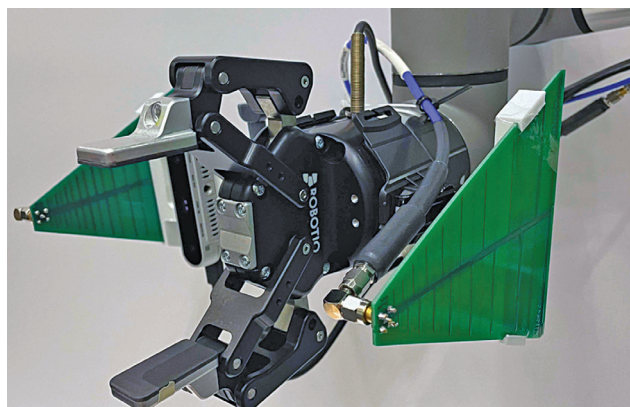
Ultrathin, freely-folding QLED display developed

Quantum dot light-emitting diode (QLED), which employs quantum dots as a light-emitting material, is known to have outstanding electroluminescence properties. Since it does not require any bulky components such as backlight units, QLED displays can potentially be manufactured into an ultrathin form factor. Now a research team led by Seoul National University has developed a foldable variant of the ultrathin QLED, inspired by the ancient art of paper folding known as origami. It is expected this technology can provide unprecedented opportunities for next-generation electronics with user-customised form factors with complex structures, as well as allowing for dynamic three-dimensional display of visual information.

Fabrication of 3D foldable QLEDs that can be folded freely like paper (Credit: www.ibs.re.kr/cop)

Robot that locates and retrieves misplaced items

Researchers at MIT have created a robotic system that can quickly search misplaced items and give them back to you. The system, RFusion, is a robotic arm with a camera and radio frequency (RF) antenna attached to its gripper. It fuses signals from the antenna with visual input from the camera to locate and retrieve an item, even if the item is buried under a pile and completely out of view. Using machine learning, the robotic arm automatically zeroes-in on the object's exact location, moves the items on top of it, grasps the object and verifies that it picked up the right thing. The camera, antenna, robotic arm, and AI are fully integrated, so RFusion can work in any environment without requiring a special set up.



Researchers at MIT have developed a fully-integrated robotic arm that fuses visual data from a camera and radio frequency information from an antenna to find and retrieve objects, even when they are buried under a pile and fully out of view (Credit: <https://news.mit.edu>)